

# E-Commerce Payment Gateway - User Workflow & Journey Map

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Version 1.0

## 1. Customer Checkout Flow

Workflow WF-ECO-001: Standard Card Payment Checkout

Step 1: Customer clicks 'Pay Now' on merchant site -> merchant site creates PaymentIntent via ECO API with amount, currency, and merchant\_id.

Step 2: ECO API returns a client\_secret token -> merchant site loads the hosted checkout iframe using client\_secret.

Step 3: Customer enters card details into hosted checkout form -> client-side validation fires on blur for each field.

Step 4: Customer clicks 'Pay' button -> hosted checkout sends encrypted card data to ECO Tokenization Service.

Step 5: Tokenization Service returns a payment\_token (within 50ms) -> checkout submits payment\_token + client\_secret to ECO Payment Processor.

Step 6: Payment Processor checks Fraud Engine (risk score returned within 100ms). If score > 85, DECLINE and display 'Payment declined for security reasons' to customer. Go to Step 10.

Step 7: If 3DS enrolled card, redirect to issuer 3DS page. On success, ECI indicator returned. On failure, DECLINE with 'Authentication failed.' Go to Step 10.

Step 8: Payment Processor sends authorization to acquiring bank via card network (Visa/MC). Authorization response within 200ms.

Step 9: If approved, PaymentIntent status updated to 'authorized'. Webhook 'payment.success' fired to merchant. Customer sees 'Payment Successful' confirmation screen.

Step 10: If declined at any step, PaymentIntent status updated to 'failed'. Webhook 'payment.failed' fired to merchant. Error displayed on checkout form with retry option.

Decision Point DP-1: If transaction amount > merchant's auto-capture threshold, payment stays in 'authorized' state for merchant to manually capture. Otherwise, auto-capture triggers and settlement is queued.

## 2. Merchant Onboarding Flow

Workflow WF-ECO-002: New Merchant Self-Service Onboarding

Step 1: Merchant visits onboarding page and fills business registration form (company name, registration number, country, website URL).

Step 2: System validates registration number against business registry API. If invalid, inline error shown.

Step 3: Merchant uploads KYC documents (certificate of incorporation, government ID of directors, proof of address).

Step 4: System runs automated document verification using Onfido API. Status: 'Under Review' for 24-48 hours.

Step 5: On KYC approval, merchant enters bank account details for settlement (IBAN or routing/account number).

Step 6: System sends micro-deposits (2 small amounts) to bank account for verification.

Step 7: Merchant confirms micro-deposit amounts -> bank account verified.

Step 8: System auto-provisions sandbox API keys and test environment. Merchant receives email with onboarding completion and API docs link.

Alternate Path AP-1: If KYC is rejected, merchant receives email with rejection reason and a link to re-upload documents.

Alternate Path AP-2: If micro-deposit verification fails 3 times, case escalated to manual review.

### 3. Refund Processing Flow

Workflow WF-ECO-003: Merchant-Initiated Refund

Step 1: Merchant navigates to transaction detail on dashboard, clicks 'Refund' button.

Step 2: Refund dialog appears. Merchant selects 'Full Refund' or 'Partial Refund'. For partial, enters refund amount (validated against original amount).

Step 3: Merchant adds refund reason (dropdown: Customer Request, Duplicate, Fraudulent, Product Issue, Other) and optional note.

Step 4: Merchant clicks 'Confirm Refund'. System validates sufficient settlement balance.

Step 5: Refund request queued in settlement engine. Status: 'Refund Initiated'. Webhook 'refund.initiated' fired.

Step 6: Settlement engine processes refund within 24 hours, reversing the original capture.

Step 7: Refund completed. Customer's card credited. Status: 'Refunded'. Webhook 'refund.completed' fired.

Error Path EP-1: If insufficient settlement balance, refund is held with status 'Pending Balance' and merchant notified.

### 4. Subscription Lifecycle Flow

Workflow WF-ECO-004: Recurring Billing Lifecycle

Step 1: Customer subscribes via merchant site. Merchant creates Subscription via API with plan\_id, payment\_token, and billing\_cycle.

Step 2: First payment charged immediately. If successful, Subscription status: 'Active'. Webhook 'subscription.created' fired.

Step 3: On each billing cycle date, system automatically charges customer's saved card.

Step 4: 3 days before charge, customer receives email notification: 'Your subscription will renew on [date] for [amount].'

Step 5: If charge fails, retry on days 1, 3, 5, 7 after initial failure.

Step 6: If all retries fail, Subscription status changes to 'Past Due'. Merchant and customer notified.

Step 7: After 7 more days in Past Due, Subscription auto-cancelled. Webhook 'subscription.cancelled' fired.

Decision Point DP-2: Customer can self-service cancel or pause subscription via merchant's portal, which calls ECO Subscription API.